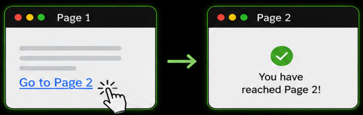
Hitesh Rathore (HR) Notes
LINKEDIN LINK:
https://www.linkedin.com/in/hitesh-rathore-835279305/

Anchor Tag :

The `<a>` tag is called the Anchor Tag.
It is used to create hyperlinks in HTML.



A hyperlink helps users:

- move from one page to another
- open files
- open images
- open email app
- call phone numbers
- jump inside same page
- download files
- navigate websites

Basic Syntax

```
<a href="URL">Text</a>
```

Where:
href = destination URL | Text = clickable text

e.g.

```
<a href="https://www.google.com">Visit Google</a>
```

Anchor Tag (`<a>`) | URL / Link (Destination) | Link Text (Clickable Text)

How it Works?

User sees the clickable text (Visit Google). → User clicks on the text. → Browser opens the given URL in a new page or same tab.

Tip: You can use any valid URL (website, file path, email, phone number, section id, etc.)

Note:

ANCHOR TAG CONTENT

```
<a href="URL">Content can be anything</a>
```

Link / Destination (URL) | Clickable Content (can be anything)

KEY POINT

The content inside the `<a>` tag is clickable.
It can be text, image, icon, button,
o any HTML element.

EXAMPLES

`<a href="about.html">About Us</a>`	Text
`<a href="home.html"><img src="home.png" alt="Home"></a>`	Image
`<a href="contact.html"><button>Contact</button></a>`	Button
`<a href="https://google.com"><i class="fa fa-google"></i> Google</a>`	Icon + Text
`<a href="docs.pdf">🗎 Download Document (PDF)</a>`	Icon + Text

VISUAL REPRESENTATION

Text Link

```
<a href="page.html">
  Visit Page
</a>
```

Clickable text
Visit Page

Image Link

```
<a href="page.html">
 <img src="img.png">
</a>
```

Clickable image

Button Link

```
<a href="page.html">
 <button>Click Me</button>
</a>
```

Clickable button
Click Me

Icon Link

```
<a href="page.html">
 <i class="fa fa-home"></i>
</a>
```

Clickable icon

Mixed Content

```
<a href="page.html">
 <i class="fa fa-download"></i>
   Download Now
</a>
```

Icon + text
Download Now

HOW IT WORKS?

User sees the clickable content. → User clicks on the content. → Browser opens the URL in a new page or same tab. → User is navigated to the destination.

TIP: You can put any valid content inside the `<a>` tag. Entire content will act as a clickable link.

MAIN PARTS OF ANCHOR TAG

```
<a href="https://youtube.com">YouTube</a>
```

Part	Meaning
`<a>`	Anchor tag
`href`	Hyper Reference (destination link)
`https://youtube.com`	URL
YouTube	Clickable text

What is href?

href means:
HYPER REFERENCE

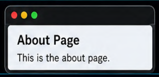
It tells browser:
"Where should I go after click?"

ANCHOR TAG EXAMPLES

①

```
<a href="about.html">About Page</a>
```

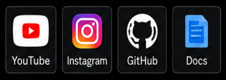
→ Clicking link opens about.html.


About Page
This is the about page.

②

```
<a href="https://www.google.com">Google</a>
```

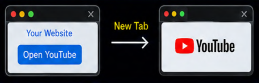
- Open YouTube
- Open Instagram
- Open GitHub
- Open Documentation



③

```
<a href="https://youtube.com" target="_blank">
  Open YouTube
</a>
```

→ Opens YouTube in a new tab.



IMPORTANT TARGET VALUES

Value	Meaning	Visual Example
`_self`	open in same tab (default)	Page A → Stays in same tab Page B
`_blank`	open in new tab	Page A → New Tab Opens Page B
`_parent`	open in parent frame	Parent Page (Frame) Page B ↑
`_top`	open in full body	Page A (in Frame) Replaces Entire Page → Page B (Full Page)

HTML LINK EXAMPLES

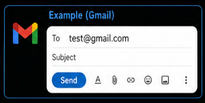
1 EMAIL LINK

```html
<a href="mailto:test@gmail.com">Send Email</a>
```

What happens?

→ Clicking this link opens the email client with the given email address.

Example (Gmail)



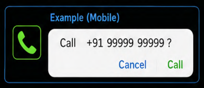
2 PHONE CALL LINK

```html
<a href="tel:+919999999999">Call Now</a>
```

What happens?

→ Clicking this link calls the given phone number directly (on mobile).

Example (Mobile)



3 DOWNLOAD FILE

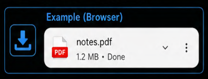
Using download attribute.

```html
<a href="notes.pdf" download>Download Notes</a>
```

What happens?

→ Browser downloads file instead of opening it.

Example (Browser)



4 WHATSAPP LINK

```html
<a href="https://wa.me/919999999999">Chat on WhatsApp</a>
```

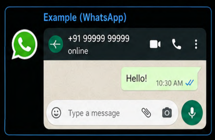
What happens?

→ Clicking this link opens a chat with the given number on WhatsApp.

Applications:

- customer support
- business inquiry

Example (WhatsApp)



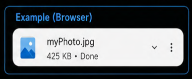
5 DOWNLOAD WITH CUSTOM NAME

```html
<a href="image.jpg" download="myPhoto">Download Image</a>
```

What happens?

→ Downloaded file name becomes: `myPhoto.jpg`

Example (Browser)



COMMON APPLICATIONS

- Email Contact
- Call Support
- Chat on WhatsApp
- Download Files

Note: These are examples of anchor `<a>` tag usage with different `href` values and attributes.

JUMP TO SECTION IN SAME PAGE

1 Step 1: Create ID

Add an `id` to the target section.

```html
<h1 id="top">Top Section</h1>
```

→ Here, `id="top"` is the target of the link.

2 Step 2: Link to ID

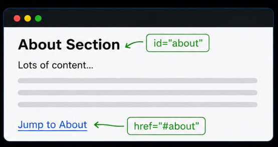
Use `#` followed by the `id` value in `href`.

```html
<a href="#top">Go to Top</a>
```

→ Clicking this link will jump to the element with `id="top"`.

Example:

```html
<h1 id="about">About Section</h1>
<p>Lots of content...</p>
 ⋮
<a href="#about">Jump to About</a>
```



→ Clicking "Jump to About" will scroll the page to the "About Section".

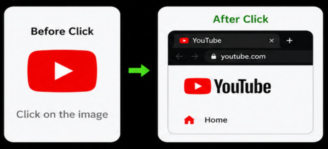
Q) How can we open a YouTube channel when clicking on an image?

Answer:

```html
<a href="https://youtube.com">
   <img src="logo.png" alt="logo">
</a>
```

Link (URL) — Clickable Image

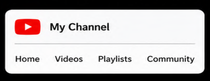
Live Preview (What happens)



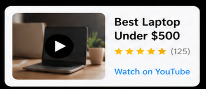
→ When the image is clicked, the link opens.

Applications:

1. Logo Navigation
Clicking on the logo takes users to your YouTube channel.



2. Product Cards
Use YouTube thumbnails as clickable images.



3. Gallery Websites
Open YouTube videos from image galleries.

Tip: You can replace the image (`logo.png`) with any custom image or thumbnail.

BUTTON LIKE LINK

```html
<a href="login.html">Login</a>
```

APPLICATIONS:

- **Signup Button**
  Link to signup page.
- **Buy Now Button**
  Link to purchase page.
- **Login Button**
  Link to login page.

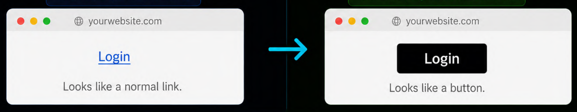
HOW IT WORKS (CSS)

Anchor can be styled like button using CSS.

```css
a {
  background: black;
  color:  white;
  padding:  10px;
}
```

- background: black; → Black background
- color: white; → White text color
- padding: 10px; → Space inside button

BEFORE (Default Link) → **AFTER (Styled Like Button)**



Tip: You can customize more: border-radius, font-size, hover effects, etc.

ANCHOR TAG ATTRIBUTES

```html
<a href="URL"  target="_blank"  title="Text">Content</a>
```

`href="URL"` → Attributes; `target="_blank"` → Attributes; `title="Text"` → Attributes; `Content` → Clickable Content

1 href

Destination link.

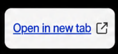
```html
<a href="https://google.com">Google</a>
```

Click to visit Google

2 target

Controls opening behavior.

```html
<a href="page.html" target="_blank">
   Open in new tab
</a>
```



Opens in a new tab

3 title

Tooltip text.

```html
<a href="page.html" title="Open About Page">
   About
</a>
```

About
Open About Page

Hover shows message

4 download

Downloads file.

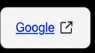
```html
<a href="file.pdf" download>Download</a>
```

Click to download the file

5 rel

It tells browser about the relationship between current page and the linked page.

```html
<a href="https://google.com"
   target="_blank"
   rel="noopener noreferrer">
   Google
</a>
```



Opens in new tab (secure)

`rel="noopener noreferrer"` improves security when using `target="_blank"`.

NOTE: You can use multiple attributes together.

```html
<a href="docs.pdf" download target="_blank" title="Download File">Download</a>
```

LINK TEXT BEST PRACTICES

Use clear, descriptive link text that tells users where the link will take them.

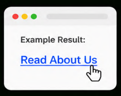
GOOD PRACTICE

```
<a href="about.html">Read About Us</a>
```

→ Clear and descriptive link text.

Why it's good:

- ✓ Users know what the link will do.
- ✓ Helps with accessibility (screen readers).
- ✓ Improves SEO and user trust.


Example Result:

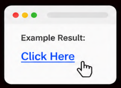
BAD

```
<a href="about.html">Click Here</a>
```

→ Unclear and unhelpful link text.

Why it's bad:

- ✗ Users don't know where it goes.
- ✗ Not accessible for screen readers.
- ✗ Poor for SEO and usability.


Example Result:

WHY IT MATTERS

Better User Experience
Users quickly understand where the link will take them.

Accessibility
Screen readers rely on meaningful link text.

SEO Friendly
Descriptive links help search engines understand content.

Builds Trust
Clear links make your website look professional and reliable.

TIP: Always write link text that describes the destination, not the action. ✓ **Good:** "View Pricing", "Download Guide" ✗ **Avoid:** "Click Here", "Read More"

rel ATTRIBUTE IN HTML

The rel attribute tells the browser:
"What is the relationship between the current page and the linked page?"

Basic syntax:

```
<a href="URL"  rel="value"> Link</a>
```

- `href="URL"`: Destination (Where to go)
- `rel="value"`: Relationship (What is the relation)
- `Link`: Clickable text (Visible to users)

Why rel Was Introduced

→ `href` tells: where to go
→ `target` tells: how to open the page

Example:

```
<a href="https://example.com"
   target="_blank" rel="noopener">
   Visit Example
</a>
```

- Where to go
- How to open
- What is the relationship

Key Takeaway:

rel provides extra information about the link.
It helps with SEO, security, accessibility, and better user experience.

SEO	Security	Accessibility	Better UX
Helps search engines	Protects users from risks	Helps assistive technologies	Improves user experience

So HTML introduced: `rel=""`

To describe the relationship between the current page and the linked page.

Most Common Usage

Usually used with:

`target="_blank"`

Example

```
<a href="https://google.com"
  target="_blank"
  rel="no opener noreferrer">
  Google
</a>
```

- Where to go
- How to open
- What is the relationship
- Clickable text

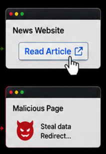
1. SECURITY PROBLEM
(window.opener)

When a page opens in a new tab using:

`target="_blank"`

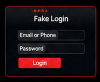
the new page gets access to:

`window.opener`



A malicious website can:

- redirect original page
- show fake login pages
- perform phishing attacks

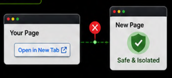


SOLUTION → noopener

`rel="noopener"`

It:

- ✓ blocks window.opener
- ✓ prevents new page from controlling old page



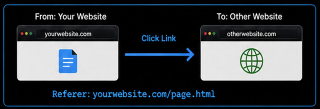
Meaning:
better security

Your Data is Safe

2. WHAT IS NOREFERRER?

`rel="noreferrer"`

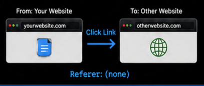
Normally browser sends:



Referer: yourwebsite.com/page.html

noreferrer:

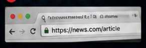
- ✓ hides that information (no referer sent)
- ✓ improves privacy



Referer: (none)

REAL WORLD EXAMPLES

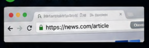
✗ Without noreferrer



Referer sent:
`https://myblog.com/post-about-tech`

The other website knows where the user came from.

✓ With noreferrer



Referer sent:
`(none)`

The other website does not know your source.

Best Practice
Use both `target="_blank"` and `rel="noopener noreferrer"` →

```
<a href="https://example.com"
  target="_blank"  rel="noopener noreferrer">
  Visit Example</a>
```

Safe Links. Happy Users.

<img> TAG IN HTML

The `<img>` tag is used to display images on a webpage.

BASIC SYNTAX

```
<img src="image.jpg" alt="Image Description">
```

<img> IS A:

- self-closing tag
- void element

THAT MEANS:

`<img></img>`	VS	`<img>`
✗ No closing tag is needed.		✓ No closing tag is needed.

HOW BROWSER UNDERSTANDS <img>

1. **HTML File**
   Browser receives the HTML file.
2. **HTML Parser**
   Browser parses the HTML.
3. **<img> Token Found**
   Parser finds the <img> tag.
4. **DOM Node Created**
   Browser creates <img> element in the DOM.
5. **src Extracted**
   Browser reads the src value.
6. **Network Request**
   Browser sends request for the image file.
7. **Image Bytes Received**
   Server sends image data (bytes).
8. **Image Decoder**
   Browser decodes the image bytes (JPG/PNG/WEBP etc.).
9. **Rendering Engine**
   Browser calculates layout and position.
10. **Pixels Painted on Screen**
    Browser paints the image pixels on the screen.

MOST IMPORTANT ATTRIBUTE → src

src means:
source

It tells browser:
"Where is the image located?"

```
<img src="image.jpg" alt="Image Description">
```

IMPORTANT ATTRIBUTE → alt

```
<img src="cat.jpg" alt="Cute Cat">
```

alt means:
alternative text

1. USED WHEN:

- To embed an image in an HTML document.
- When the image is essential for content or understanding.
- When the image fails to load.
  - → Screen readers read the alt text.
  - → Browsers show the alt text (if available).
  - → Users still understand the purpose even without the image.

Image loads

Image fails to load

Mountain landscape with sun

2. IF IMAGE NOT FOUND:

- If the image resource cannot be loaded (404, wrong path, server down, etc.).
  - → Browser displays the alt text.
  - → Screen readers read the alt text.

Example in browser:

Image not found

Alt text: "Product image"

3. BROWSER SHOWS:

- The alt text (if provided) in place of the missing image.
- Or a broken image icon (if alt is not provided).

With alt text	Without alt text
Summer beach vacation	

4. WHY ALT IS IMPORTANT:

- **Accessibility:** Helps visually impaired users via screen readers.
- **SEO:** Search engines understand image content through alt text.
- **Better UX:** Provides information if image cannot be displayed.

Screen Reader → Reading... "Delicious pizza with cheese"

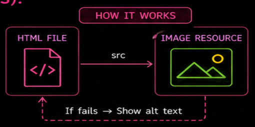
5. APPLICATIONS:

- Accessibility compliance (WCAG).
- SEO optimization.
- Better user experience in various conditions (slow internet, errors, etc.).

Accessibility | SEO | Works Offline / Slow Internet | Error Handling

6. FIRST PRINCIPLE (TECHNICAL BASIS):

- An <img> tag does NOT contain the image itself.
- It only provides the **source (src)** — a reference to the image resource.
- When the browser parses the <img> tag:
  - → It requests the image from the given source.
  - → If the request fails or the resource is unavailable, the browser cannot display the image.
  - → The **alt** attribute provides alternative text, ensuring the user still gets the information.



1. WIDTH AND HEIGHT

```
<img src="cat.jpg" width="300" height="200">
```

WHAT IT DOES:
The `width` and `height` attributes specify the display size of the image in pixels.

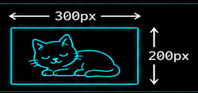
BEST PRACTICE USING CSS:

```
<img src="image.jpg" style="width:300px;">
```

Using CSS (like `width:300px;)` gives more flexibility, better control, and cleaner code.

IMPORTANT:
Width and height only control the space the image takes.
They do not change the actual image file size.

EXAMPLE:



EFFECT:
The image will be displayed in a 300px wide and 200px tall space.

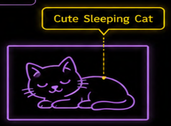
2. TITLE ATTRIBUTE

```
<img src="cat.jpg" title="Cute Sleeping Cat">
```

WHAT IT DOES:
The `title` attribute provides extra information about the image.

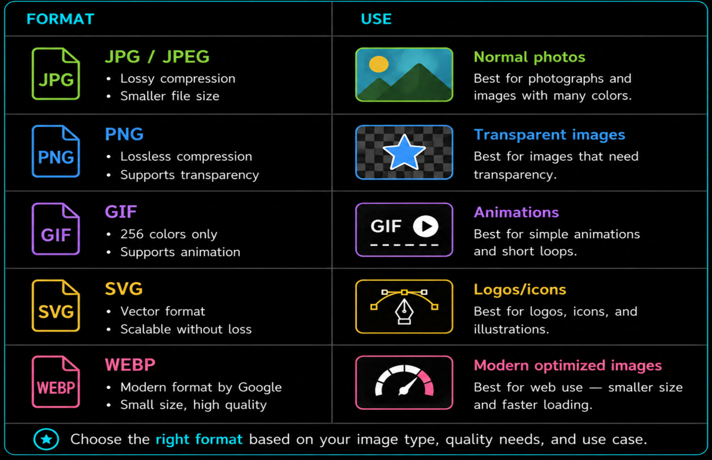
WHEN MOUSE HOVERS ON IMAGE:
The title text appears as a tooltip (small popup).

BROWSER BEHAVIOR:
The tooltip is shown by the browser automatically. Style and position may vary slightly across browsers.



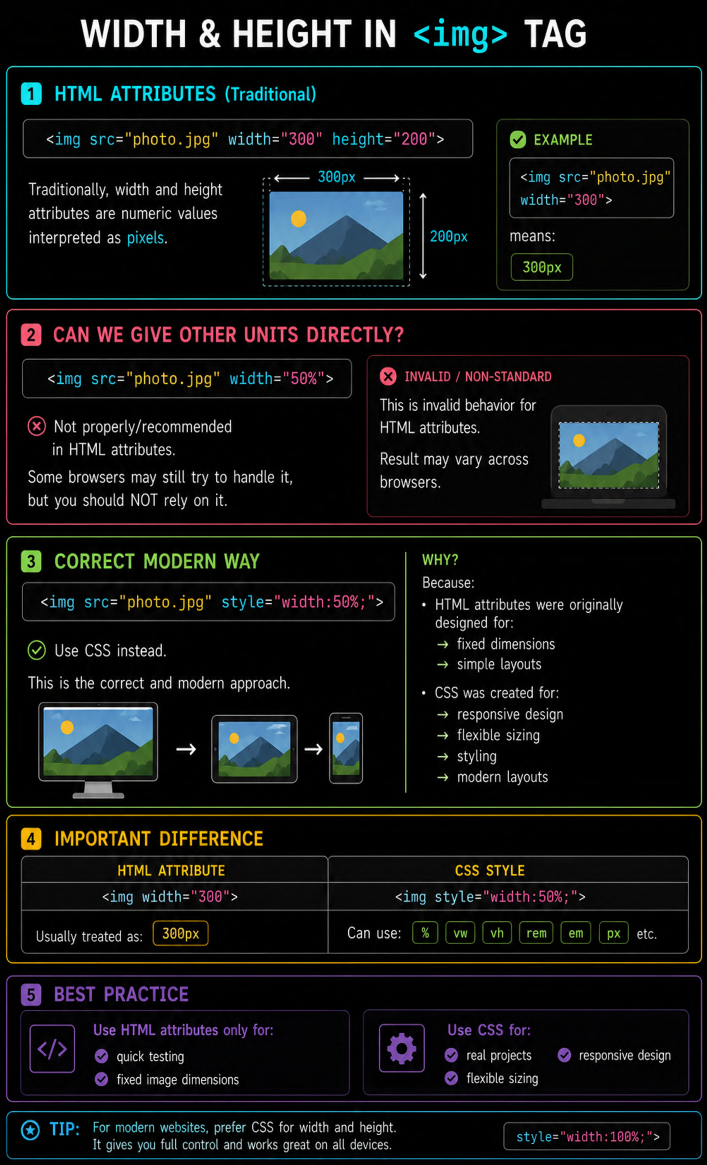
Use `title` attribute to provide helpful information to users when they interact with the image.

IMAGE FORMATS



FORMAT	USE
JPG / JPEG • Lossy compression • Smaller file size	**Normal photos** Best for photographs and images with many colors.
PNG • Lossless compression • Supports transparency	**Transparent images** Best for images that need transparency.
GIF • 256 colors only • Supports animation	**Animations** Best for simple animations and short loops.
SVG • Vector format • Scalable without loss	**Logos/icons** Best for logos, icons, and illustrations.
WEBP • Modern format by Google • Small size, high quality	**Modern optimized images** Best for web use — smaller size and faster loading.

Choose the **right format** based on your image type, quality needs, and use case.



WIDTH & HEIGHT IN <img> TAG

1 HTML ATTRIBUTES (Traditional)

`<img src="photo.jpg" width="300" height="200">`

Traditionally, width and height attributes are numeric values interpreted as pixels.

300px
200px

EXAMPLE

`<img src="photo.jpg" width="300">`

means:

`300px`

2 CAN WE GIVE OTHER UNITS DIRECTLY?

`<img src="photo.jpg" width="50%">`

☒ Not properly/recommended in HTML attributes.

Some browsers may still try to handle it, but you should NOT rely on it.

INVALID / NON-STANDARD

This is invalid behavior for HTML attributes.

Result may vary across browsers.

3 CORRECT MODERN WAY

`<img src="photo.jpg" style="width:50%;">`

✓ Use CSS instead.

This is the correct and modern approach.

WHY?

Because:

- HTML attributes were originally designed for:
  - → fixed dimensions
  - → simple layouts
- CSS was created for:
  - → responsive design
  - → flexible sizing
  - → styling
  - → modern layouts

4 IMPORTANT DIFFERENCE

HTML ATTRIBUTE	CSS STYLE
`<img width="300">`	`<img style="width:50%;">`
Usually treated as: `300px`	Can use: `%` `vw` `vh` `rem` `em` `px` etc.

5 BEST PRACTICE

Use HTML attributes only for:

- ✓ quick testing
- ✓ fixed image dimensions

Use CSS for:

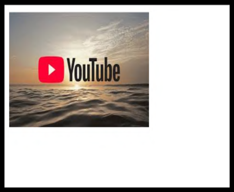
- ✓ real projects
- ✓ responsive design
- ✓ flexible sizing

TIP: For modern websites, prefer CSS for width and height. It gives you full control and works great on all devices.

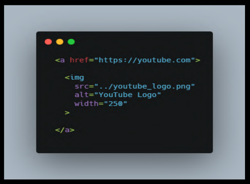
`style="width:100%;">`

Now it's time to implement these concepts practically:

Practice Task 1

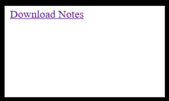


Code →



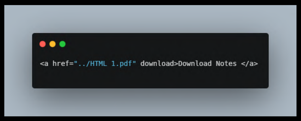
```
<a href="https://youtube.com">

  <img
    src="../youtube_logo.png"
    alt="YouTube Logo"
    width="250"
  >

</a>
```

Practice Tast 2



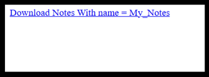
Download Notes

Code →



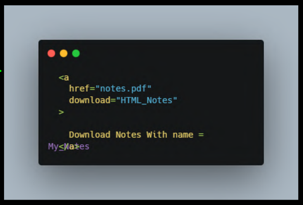
```
<a href="../HTML 1.pdf" download>Download Notes </a>
```

Practice Task 3



Download Notes With name = My_Notes

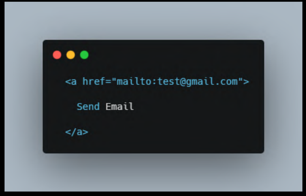
Code →



```
  <a
    href="notes.pdf"
    download="HTML_Notes"
  >

    Download Notes With name =
My_</a>tes
```

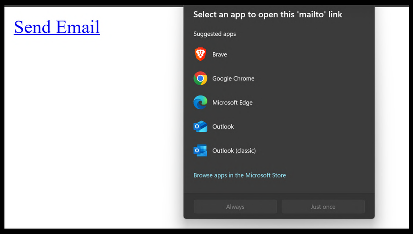
Practice Tast 4

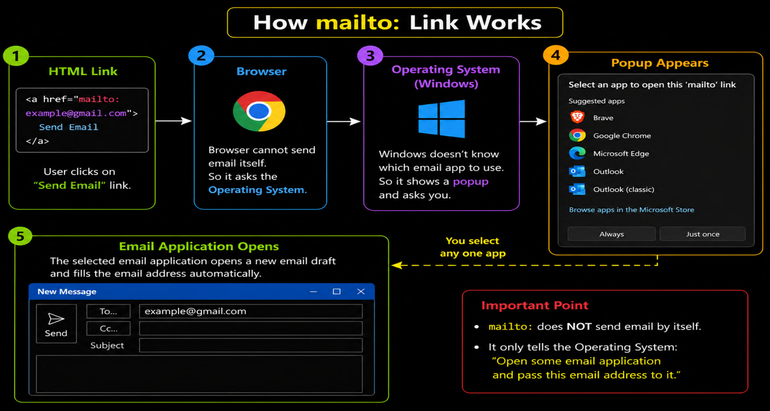
Code



```
<a href="mailto:test@gmail.com">

  Send Email

</a>
```

On click it will show Popup





How mailto: Link Works

1 HTML Link

```
<a href="mailto:
example@gmail.com">
  Send Email
</a>
```

User clicks on "Send Email" link.

2 Browser

Browser cannot send email itself. So it asks the Operating System.

3 Operating System (Windows)

Windows doesn't know which email app to use. So it shows a popup and asks you.

4 Popup Appears

You select any one app

5 Email Application Opens

The selected email application opens a new email draft and fills the email address automatically.

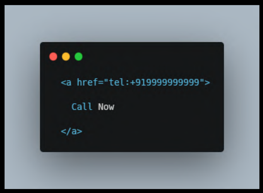
To...: example@gmail.com

Important Point

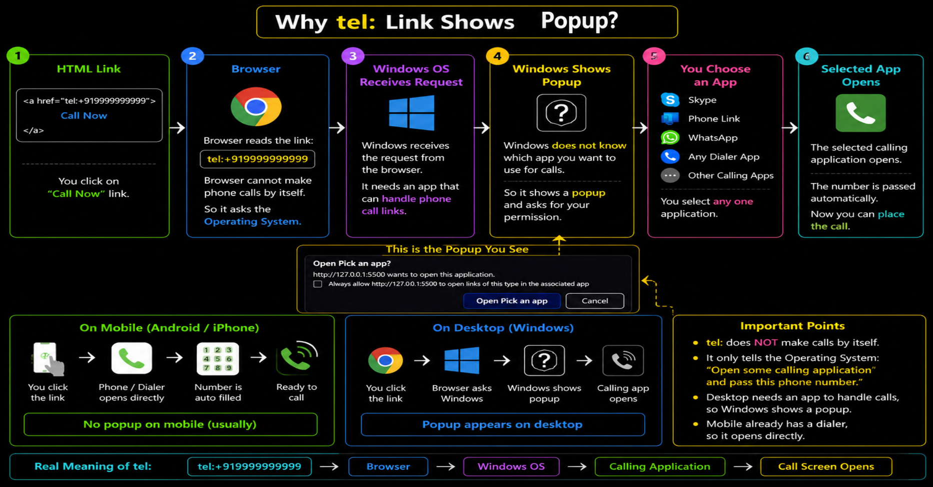
- `mailto:` does **NOT** send email by itself.
- It only tells the Operating System:
  "Open some email application
  and pass this email address to it."

Practice Task 5

Code



```
<a href="tel:+919999999999">

  Call Now

</a>
```



Why tel: Link Shows Popup?

1 HTML Link

```
<a href="tel:+919999999999">
  Call Now
</a>
```

You click on "Call Now" link.

2 Browser

Browser reads the link:

tel:+919999999999

Browser cannot make phone calls by itself.

So it asks the Operating System.

3 Windows OS Receives Request

Windows receives the request from the browser.

It needs an app that can handle phone call links.

4 Windows Shows Popup

Windows does not know which app you want to use for calls.

So it shows a popup and asks for your permission.

5 You Choose an App

- Skype
- Phone Link
- WhatsApp
- Any Dialer App
- Other Calling Apps

You select any one application.

6 Selected App Opens

The selected calling application opens.

The number is passed automatically.

Now you can place the call.

This is the Popup You See

Open Pick an app?

http://127.0.0.1:5500 wants to open this application.

☐ Always allow http://127.0.0.1:5500 to open links of this type in the associated app

On Mobile (Android / iPhone)

You click the link → Phone / Dialer opens directly → Number is auto filled → Ready to call

No popup on mobile (usually)

On Desktop (Windows)

You click the link → Browser asks Windows → Windows shows popup → Calling app opens

Popup appears on desktop

Important Points

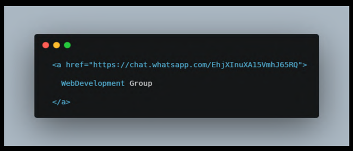
- tel: does NOT make calls by itself.
- It only tells the Operating System:
  "Open some calling application"
  and pass this phone number."
- Desktop needs an app to handle calls, so Windows shows a popup.
- Mobile already has a dialer, so it opens directly.

Real Meaning of tel: tel:+919999999999 → Browser → Windows OS → Calling Application → Call Screen Opens

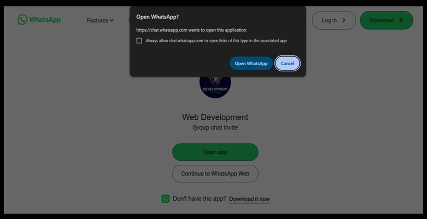
Practice Task 6

WebDevelopment Group

Code



```
<a href="https://chat.whatsapp.com/EhjXInuXA15VmhJ65RQ">

  WebDevelopment Group

</a>
```



Open WhatsApp?

https://chat.whatsapp.com wants to open this application.

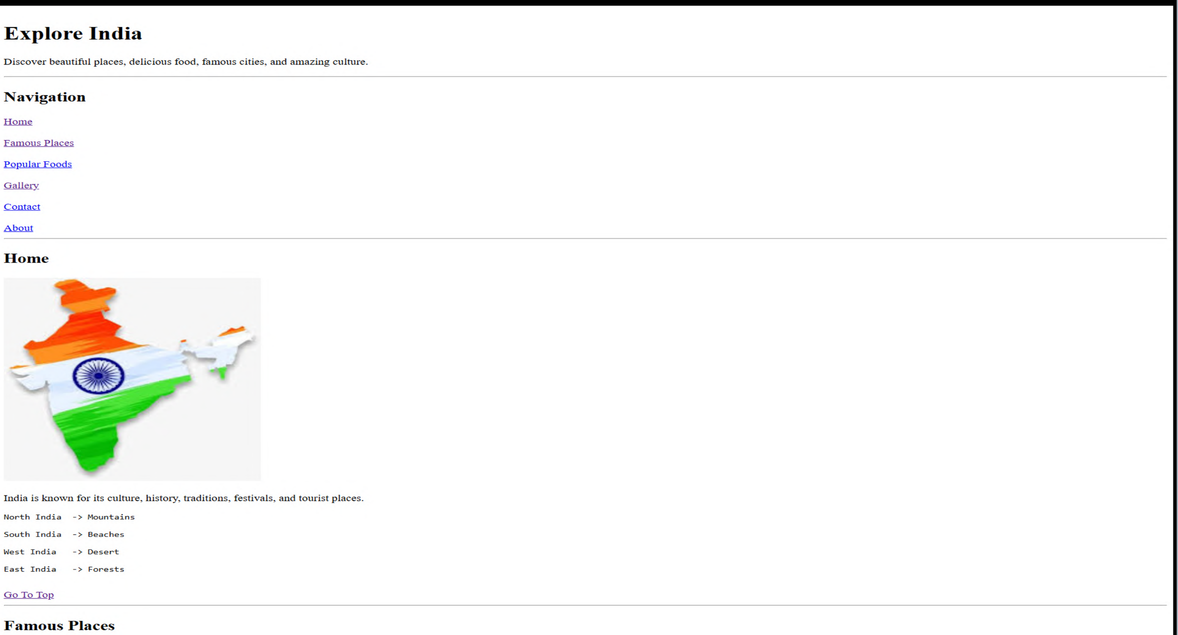
☐ Always allow chat.whatsapp.com to open links of this type in the associated app

Web Development

Group chat invite

Don't have the app? Download it now

Main Task



Explore India

Discover beautiful places, delicious food, famous cities, and amazing culture.

Navigation

Home

Famous Places

Popular Foods

Gallery

Contact

About

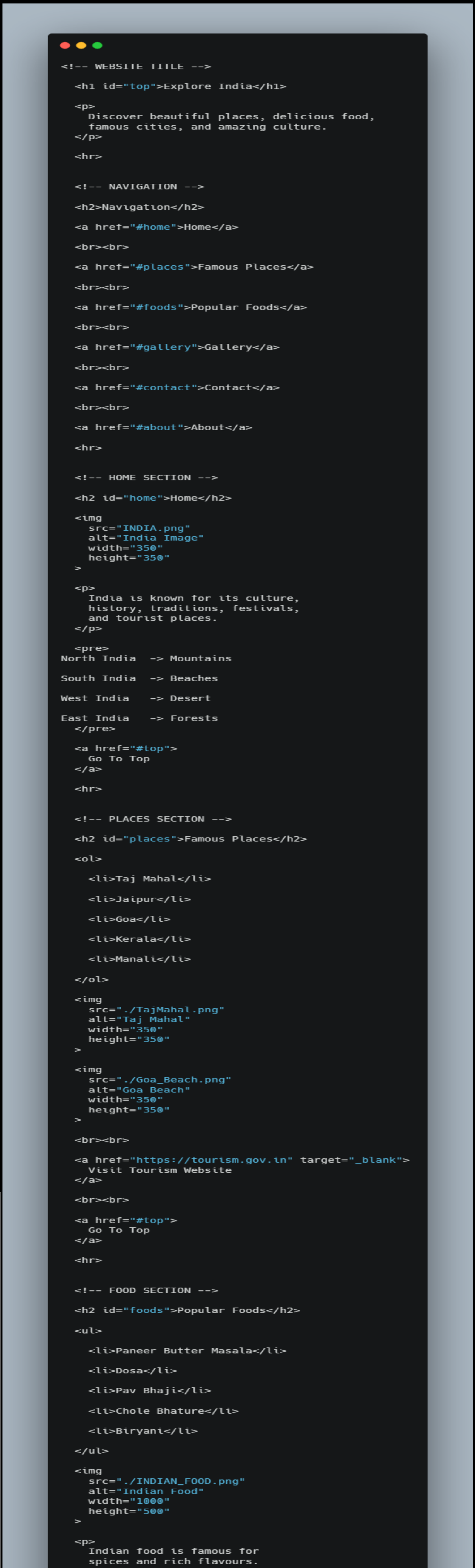
Home

India is known for its culture, history, traditions, festivals, and tourist places.

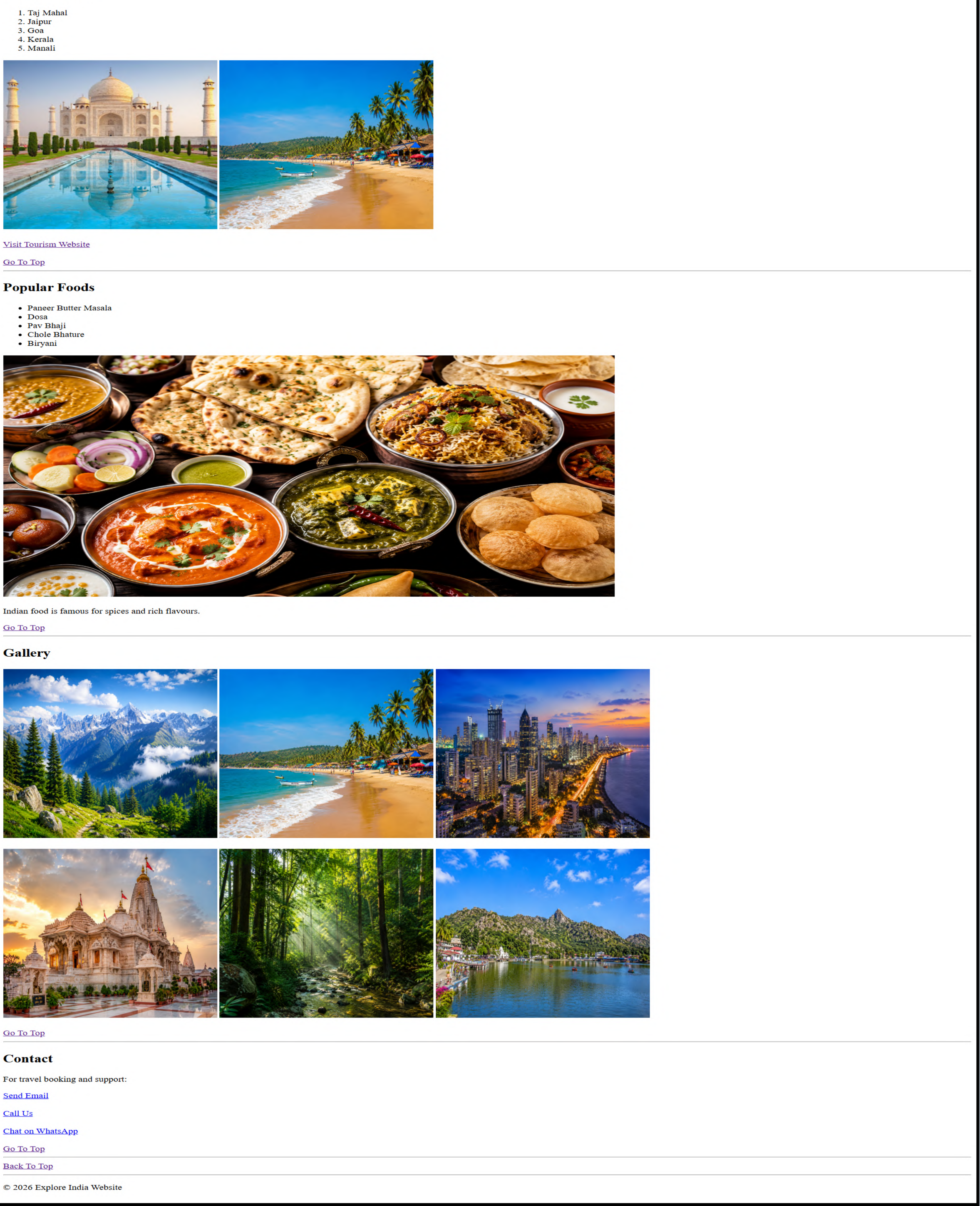
```
North India  -> Mountains
South India  -> Beaches
West India   -> Desert
East India   -> Forests
```

Go To Top

Famous Places



```
<!-- WEBSITE TITLE -->

  <h1 id="top">Explore India</h1>

  <p>
    Discover beautiful places, delicious food,
    famous cities, and amazing culture.
  </p>

  <hr>


  <!-- NAVIGATION -->

  <h2>Navigation</h2>

  <a href="#home">Home</a>

  <br><br>

  <a href="#places">Famous Places</a>

  <br><br>

  <a href="#foods">Popular Foods</a>

  <br><br>

  <a href="#gallery">Gallery</a>

  <br><br>

  <a href="#contact">Contact</a>

  <br><br>

  <a href="#about">About</a>

  <hr>


  <!-- HOME SECTION -->

  <h2 id="home">Home</h2>

  <img
    src="INDIA.png"
    alt="India Image"
    width="350"
    height="350"
  >

  <p>
    India is known for its culture,
    history, traditions, festivals,
    and tourist places.
  </p>

  <pre>
North India  -> Mountains

South India  -> Beaches

West India   -> Desert

East India   -> Forests
  </pre>

  <a href="#top">
    Go To Top
  </a>

  <hr>


  <!-- PLACES SECTION -->

  <h2 id="places">Famous Places</h2>

  <ol>

    <li>Taj Mahal</li>

    <li>Jaipur</li>

    <li>Goa</li>

    <li>Kerala</li>

    <li>Manali</li>

  </ol>

  <img
    src="./TajMahal.png"
    alt="Taj Mahal"
    width="350"
    height="350"
  >

  <img
    src="./Goa_Beach.png"
    alt="Goa Beach"
    width="350"
    height="350"
  >

  <br><br>

  <a href="https://tourism.gov.in" target="_blank">
    Visit Tourism Website
  </a>

  <br><br>

  <a href="#top">
    Go To Top
  </a>

  <hr>


  <!-- FOOD SECTION -->

  <h2 id="foods">Popular Foods</h2>

  <ul>

    <li>Paneer Butter Masala</li>

    <li>Dosa</li>

    <li>Pav Bhaji</li>

    <li>Chole Bhature</li>

    <li>Biryani</li>

  </ul>

  <img
    src="./INDIAN_FOOD.png"
    alt="Indian Food"
    width="1000"
    height="500"
  >

  <p>
    Indian food is famous for
    spices and rich flavours.
```



1. Taj Mahal
2. Jaipur
3. Goa
4. Kerala
5. Manali

Visit Tourism Website

Go To Top

Popular Foods

- Paneer Butter Masala
- Dosa
- Pav Bhaji
- Chole Bhature
- Biryani

Indian food is famous for spices and rich flavours.

Go To Top

Gallery

Go To Top

Contact

For travel booking and support:

Send Email

Call Us

Chat on WhatsApp

Go To Top

Back To Top

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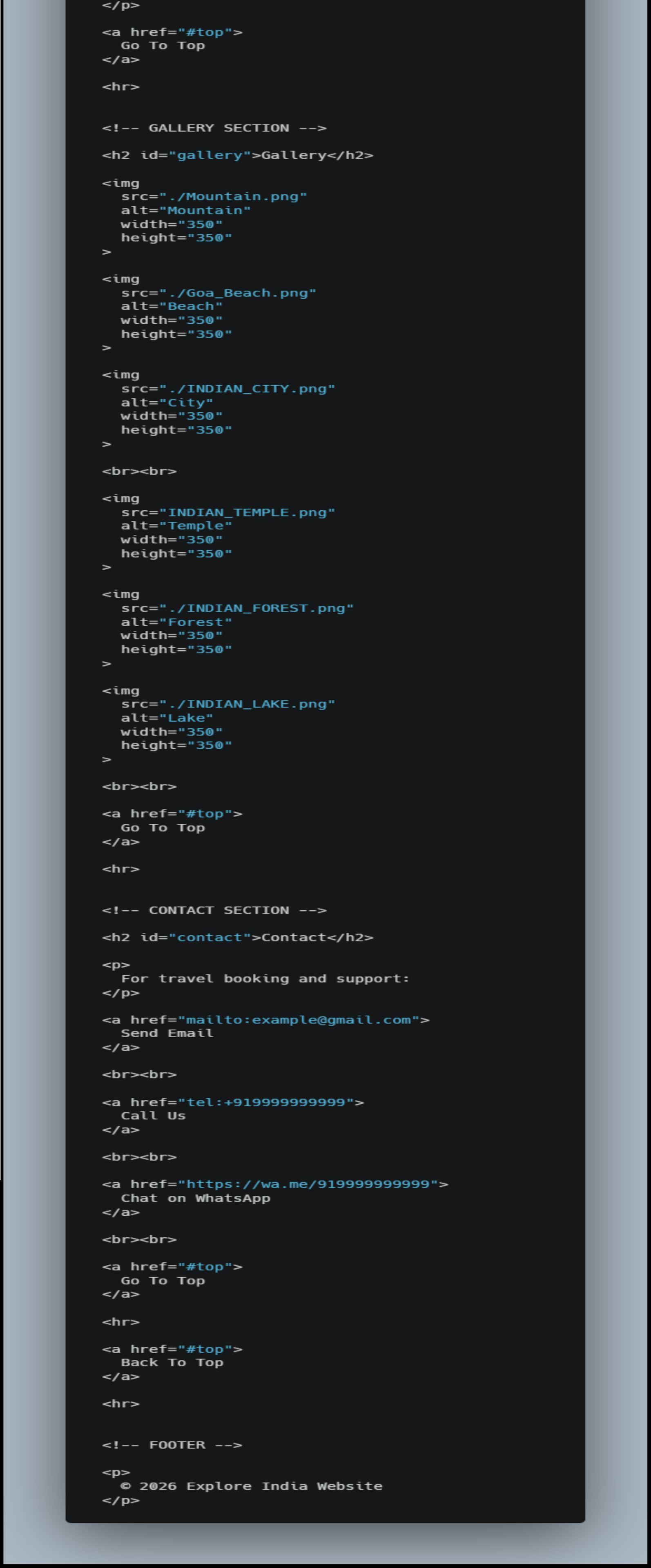

```
</p>

<a href="#top">
  Go To Top
</a>

<hr>


<!-- GALLERY SECTION -->

<h2 id="gallery">Gallery</h2>

<img
  src="./Mountain.png"
  alt="Mountain"
  width="350"
  height="350"
>

<img
  src="./Goa_Beach.png"
  alt="Beach"
  width="350"
  height="350"
>

<img
  src="./INDIAN_CITY.png"
  alt="City"
  width="350"
  height="350"
>

<br><br>

<img
  src="INDIAN_TEMPLE.png"
  alt="Temple"
  width="350"
  height="350"
>

<img
  src="./INDIAN_FOREST.png"
  alt="Forest"
  width="350"
  height="350"
>

<img
  src="./INDIAN_LAKE.png"
  alt="Lake"
  width="350"
  height="350"
>

<br><br>

<a href="#top">
  Go To Top
</a>

<hr>


<!-- CONTACT SECTION -->

<h2 id="contact">Contact</h2>

<p>
  For travel booking and support:
</p>

<a href="mailto:example@gmail.com">
  Send Email
</a>

<br><br>

<a href="tel:+919999999999">
  Call Us
</a>

<br><br>

<a href="https://wa.me/919999999999">
  Chat on WhatsApp
</a>

<br><br>

<a href="#top">
  Go To Top
</a>

<hr>

<a href="#top">
  Back To Top
</a>

<hr>


<!-- FOOTER -->

<p>
  © 2026 Explore India Website
</p>
```